

CHICAGO DUPAGE, IL, USA

WMO: 725305

Lat: 41.914N

Lon: 88.246W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.1	-24.1	0.4	-19.2	-21.9	0.5	-16.5	12.2	-3.5	11.1	-3.3	4.2	270	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.4	23.7	31.1	23.0	29.5	22.1	25.4	30.5	24.4	29.0	23.5	27.9	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.2	28.7	22.8	18.0	27.5	22.1	17.3	26.5	79.0	30.4	74.8	29.0	71.0	27.9	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.6	DB	-24.2	34.6	3.8	1.5	-26.9	35.7	-29.1	36.6	-31.3	37.4	-34.0	38.6	(4)
(5)			WB	-24.5	27.0	3.7	0.9	-27.1	27.7	-29.3	28.2	-31.4	28.7	-34.1	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-5.2	-3.0	3.0	9.3	15.3	20.6	22.9	21.9	18.3	11.4	4.2	-2.0	(6)	
	DBStd	11.03	6.73	6.30	6.10	5.01	4.78	3.92	3.15	2.88	4.21	4.94	5.33	6.14	(7)	
	HDD10.0	1753	472	366	233	71	8	0	0	0	1	44	186	374	(8)	
	HDD18.3	3558	730	598	477	273	121	21	3	4	53	222	424	631	(9)	
	CDD10.0	1676	0	1	14	51	172	319	401	369	249	86	11	2	(10)	
	CDD18.3	441	0	0	1	3	27	91	146	116	52	6	0	0	(11)	
	CDH23.3	4034	0	0	8	44	273	874	1379	942	457	57	0	0	(12)	
	CDH26.7	1263	0	0	1	7	73	293	473	276	131	10	0	0	(13)	
(14) Wind	WSAvg	4.1	4.8	4.7	4.7	5.0	4.3	3.7	3.3	3.0	3.3	4.0	4.4	4.4	(14)	
(15) Precipitation	PrecAvg	918	48	44	57	89	106	108	96	103	77	77	65	48	(15)	
	PrecMax	1125	102	144	128	176	184	252	172	233	322	199	140	120	(16)	
	PrecMin	606	12	10	19	31	14	25	40	47	10	19	10	15	(17)	
	PrecStd	125	25	27	27	39	43	54	37	42	60	49	38	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.4	24.3	27.6	31.3	33.6	35.0	32.9	32.3	28.1	20.7	15.5	(19)	
		MCWB	10.6	11.8	17.0	18.2	21.2	24.1	25.2	23.9	22.9	20.6	14.3	13.7	(20)	
	2%	DB	9.0	12.2	19.1	24.1	28.7	32.0	32.5	31.2	30.1	24.8	17.3	12.1	(21)	
		MCWB	7.3	9.0	13.5	16.1	19.9	22.8	24.4	23.5	22.1	17.6	13.2	9.7	(22)	
	5%	DB	5.8	8.1	16.1	21.4	26.6	30.0	31.0	29.5	27.9	22.3	15.1	8.8	(23)	
		MCWB	4.0	5.4	11.6	14.5	18.8	22.1	23.6	22.4	20.5	16.7	11.9	7.1	(24)	
	10%	DB	3.3	5.2	12.7	18.5	23.9	28.0	29.2	27.9	26.2	19.8	12.5	6.2	(25)	
		MCWB	2.0	3.0	8.7	12.8	17.8	21.2	22.4	21.5	19.6	15.7	9.5	4.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.1	17.6	19.8	23.4	26.4	26.9	26.0	24.5	21.9	16.3	14.0	(27)	
		MCDB	12.8	15.3	23.2	25.9	29.0	31.9	32.1	31.3	30.2	26.9	18.4	15.2	(28)	
	2%	WB	7.6	9.2	15.0	17.3	21.5	24.5	25.7	24.8	23.0	19.5	14.3	10.2	(29)	
		MCDB	8.7	12.1	18.4	22.5	26.7	29.3	30.7	29.3	27.9	22.4	16.6	11.2	(30)	
	5%	WB	4.1	5.7	12.3	15.5	20.1	23.3	24.7	23.7	21.9	17.8	12.0	7.0	(31)	
		MCDB	5.5	8.2	15.3	20.1	24.7	28.4	29.4	27.7	26.1	21.0	14.2	8.3	(32)	
	10%	WB	2.1	3.2	8.8	13.5	18.7	22.1	23.7	22.7	20.7	15.9	9.9	4.4	(33)	
		MCDB	3.4	5.0	12.5	18.1	22.9	26.9	28.0	26.5	24.6	19.3	12.5	6.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.7	10.0	11.5	11.7	11.3	10.9	10.7	11.7	10.8	9.1	7.6	(35)	
		MCDBR	9.9	12.2	14.0	16.0	14.3	12.7	11.8	12.0	13.4	13.9	12.3	9.6	(36)	
	5% WB	MCWBR	8.4	9.4	9.5	9.7	7.6	6.3	5.7	5.8	6.5	8.3	9.2	8.3	(37)	
		MCWBR	9.5	11.4	12.5	14.0	12.5	11.2	10.6	10.2	11.4	11.3	10.8	9.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.293	0.310	0.343	0.397	0.423	0.433	0.441	0.433	0.406	0.353	0.310	0.293	(40)	
		taud	2.417	2.377	2.384	2.250	2.231	2.242	2.217	2.236	2.286	2.432	2.502	2.446	(41)	
	Edn at Noon	Edn at Noon	858	896	903	872	855	844	832	827	824	835	833	826	(42)	
		Edh at Noon	78	95	107	132	138	137	139	132	117	90	72	70	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.62	2.53	3.60	4.64	5.31	6.06	6.08	5.32	4.43	2.90	1.86	1.36	(44)	
	RadStd	RadStd	0.19	0.32	0.29	0.38	0.42	0.42	0.32	0.31	0.38	0.33	0.17	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+88	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page